RAG BASELINE VS RERANKING

Setup

- Document: GDPR, Regulation (EU) 2016/679, 88-page PDF
- 349 chunks, 200 words each, 40-word overlap (same chunks for both pipelines)
- Bi-Encoder: gemini-embedding-001, 768-dim, normalised, FAISS IndexFlatIP (exact search, inner product = cosine)
- Cross-Encoder: cross-encoder/ms-marco-MiniLM-L-6-v2
- LLM: gemini-2.5-flash, temperature 0, identical prompt, top 3 chunks as context
- Hardware: Colab CPU
- 12 stress queries (negation, exception, keyword trap, buried). Flag rule: a chunk from baseline rank 4-10 enters the new top 3. 11 of 12 flagged (only child-consent query unchanged). Flag shows the context CHANGED, not that the baseline was wrong.

1. Architecture

- A Bi-encoder embeds the query and each chunk separately , and cosine compares them
- A cross-encoder takes the query and chunks it together in one pass so attention can link the question's words to the passage.

2. Rank Shifts and Answers

- **Automated decisions:** chunk 228 moved #10→#3 and chunk 203 moved #6→#2. The baseline's #1 chunk (204) dropped to #10. Answer 2 added the contract/authorised exceptions; Answer 1 had only the general right.
- **Erasure:** chunk 62 moved #8→#3, but the key chunk 197 dropped #1→#4. Answer 1 was correct; Answer 2 was wrong. Re-ranking hurts here.
- **Religion/politics:** chunk 70 moved #10→#2, but chunk 176 dropped #3→#5. Answer 2 listed fewer exceptions.
- **Adequacy:** chunk 255 moved #7→#3, but chunk 257 dropped #2→#9. No real gain.
- **Portability:** chunk 200 moved #5→#2. Both answers were correct, and Answer 2 used the Art. 20 text.
- **Penalty:** chunk 329 moved #5→#3. The right chunk (333) was already #2, so the answers were identical.

3. Latency (points)

- **Setup:** measured on a Colab CPU over 12 queries.
- **Baseline retrieval:** about 148.5 ms on average (median 152.9 ms, max 170.2 ms).
    - Query embedding (Gemini API call): about 148.3 ms on average.
    - FAISS search for the top 10: about 0.2 ms on average.
    - Almost all baseline time is the API call, not the search.
- **Re-ranking:** adds about 1,807 ms on average (median 1,445 ms, max 4,567 ms).
    - That is about 12 times the baseline retrieval time.
    - The 4.57 s maximum came from one query (portability), and without it the average is about 1.56 s.
- **Full Pipeline 2 retrieval:** about 1,956 ms on average (baseline plus re-ranking).
- **LLM answer generation:** about 2,971 ms on average (median 2,872 ms), the same for both pipelines.
    - The LLM call is still the slowest single step.
- **What it means:** re-ranking costs noticeably more time than baseline retrieval, but it is smaller than the LLM call.
- **Caveat:** a GPU would likely lower the re-ranking time, but I didn't measure that.

4. Scaling (points)

- **Measured Cross-Encoder times:**
    - 10 chunks: about 1,385 ms
    - 50 chunks: about 8,628 ms
    - 100 chunks: about 15,338 ms
    - 349 chunks: about 54,878 ms
- **Pattern:** about 157 ms per chunk, so the cost grows linearly with the number of chunks.
- **Estimates for one query:**
    - 10,000 chunks: about 26 minutes
    - 1,000,000 chunks: about 2,620 minutes (about 44 hours)
- **Why not run it on everything:**
    - A Cross-Encoder score depends on the query and the chunk together, so nothing can be precomputed. Every new query means scoring every chunk again.
    - A Bi-Encoder embeds each chunk once, ahead of time. A query then needs one embedding plus a fast vector search (0.2 ms here).
    - Approximate-nearest-neighbour indexes keep that search quick even with millions of chunks.
- **How the two stages fit:**
    - Stage 1 is cheap and gives recall: it narrows the collection to the top 10.
    - Stage 2 is expensive and gives precision: it re-sorts only that short list.
- **Limit to mention:** if the right chunk isn't in the stage-1 top 10, re-ranking can't recover it.